Three-Flavour Quartic: Maximal 1|3 Entropy and RG-Invariant Couplings

Detailed computation note

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Abstract

For a fully symmetric quartic coupling tensor $\lambda_{ijkl} = \lambda_{(ijkl)}$ with $i, j, k, l = 1, 2, 3$, we compute the locus on which the one-leg reduced density matrix is maximally mixed, and then impose one-loop RG invariance. The result is the $O(3)$ orbit of the two-parameter cubic/tetrahedral plane

$$\lambda_{iiii} = u, \quad \lambda_{iijj} = w \ (i \neq j), \quad \text{all other components} = 0,$$

equivalently

$$\lambda_{ijkl} = w\,(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + (u - 3w)\sum_{a=1}^{3} n_i^a n_j^a n_k^a n_l^a,$$

where $\{n^1, n^2, n^3\}$ is an orthonormal frame. The $SO(3)$-symmetric line is the sublocus $u = 3w$. We prove this is the *complete* real solution set of entropy-maximality plus RG-invariance, modulo $O(3)$: the harmonic (spin-4) reduction has exactly one $O(3)$ orbit.

1 Setup and component notation

The quartic interaction is

$$V(\phi) = \tfrac{1}{4!}\lambda_{ijkl}\phi_i\phi_j\phi_k\phi_l, \qquad \lambda_{ijkl} = \lambda_{(ijkl)}, \quad i, j, k, l = 1, 2, 3. \tag{1}$$

There are $\dim \mathrm{Sym}^4(\mathbb{R}^3) = \binom{6}{4} = 15$ independent components. We write

$$\begin{aligned}
&A = \lambda_{1111},\ B = \lambda_{1112},\ C = \lambda_{1113},\ D = \lambda_{1122},\ E = \lambda_{1123},\\
&F = \lambda_{1133},\ G = \lambda_{1222},\ H = \lambda_{1223},\ I = \lambda_{1233},\ J = \lambda_{1333},\\
&K = \lambda_{2222},\ L = \lambda_{2223},\ M = \lambda_{2233},\ N = \lambda_{2333},\ P = \lambda_{3333}.
\end{aligned} \tag{2}$$

Treating the tensor as a four-party state, the unnormalised one-leg reduced matrix for the 1|3 partition is

$$R_{ij} := (\rho'_{1|3})_{ij} = \sum_{k,l,m=1}^{3} \lambda_{iklm}\lambda_{jklm}. \tag{3}$$

The normalised entropy is maximal exactly when R has three equal eigenvalues,

$$T_{ij} := R_{ij} - \tfrac{1}{3}(\mathrm{tr}\, R)\delta_{ij} = 0, \tag{4}$$

the spin-2, symmetric traceless part of R.

2 The five entropy equations

The entries of R are

$$
\begin{aligned}
R_{11} &= A^2 + 3B^2 + 3C^2 + 3D^2 + 6E^2 + 3F^2 + G^2 + 3H^2 + 3I^2 + J^2, \\
R_{12} &= AB + 3BD + 3CE + 3DG + 6EH + 3FI + GK + 3HL + 3IM + JN, \\
R_{13} &= AC + 3BE + 3CF + 3DH + 6EI + 3FJ + GL + 3HM + 3IN + JP, \\
R_{22} &= B^2 + 3D^2 + 3E^2 + 3G^2 + 6H^2 + 3I^2 + K^2 + 3L^2 + 3M^2 + N^2, \\
R_{23} &= BC + 3DE + 3EF + 3GH + 6HI + 3IJ + KL + 3LM + 3MN + NP, \\
R_{33} &= C^2 + 3E^2 + 3F^2 + 3H^2 + 6I^2 + 3J^2 + L^2 + 3M^2 + 3N^2 + P^2.
\end{aligned}
\tag{5}
$$

The full $1|3$ entropy-maximal locus is the real algebraic variety

$$
\mathcal{M}_{1|3} = \{R_{12} = R_{13} = R_{23} = 0,\ R_{11} = R_{22} = R_{33}\}, \tag{6}
$$

on which $\widehat{\rho}_{1|3} = \frac{1}{3}\mathbb{1}_3$ and $S_{1|3} = \log 3$. (The origin solves the equations but is not a normalised state.)

3 One-loop RG flow

We use the one-loop tensor beta function with the $1/4!$ normalisation,

$$
\dot{\lambda}_{ijkl} = -\lambda_{ijkl} + \kappa\,(\lambda_{ijmn}\lambda_{mnkl} + \lambda_{ikmn}\lambda_{mnjl} + \lambda_{ilmn}\lambda_{mnjk}), \tag{7}
$$

with m, n summed over $1, 2, 3$. The linear term never affects tangency to $T = 0$ (it rescales T by $-2T$); the nonlinear term decides invariance. For any polynomial $F(\lambda)$ define the Lie derivative along the RG vector field by $\mathcal{L}_\beta F := \beta_a\, \partial F/\partial\lambda_a$. The genuinely invariant entropy-maximal locus is

$$
\Sigma := \{\lambda :\ T = \mathcal{L}_\beta T = \mathcal{L}_\beta^2 T = \cdots = 0\}. \tag{8}
$$

This is stronger than first-order tangency: in the three-flavour calculation, first- and second-order tangency admit extra real branches, but direct flow integration shows those branches leave $\mathcal{M}_{1|3}$ at cubic order in RG time. The Lie-chain condition removes them (see Remark 1).

4 A closed entropy-maximal plane

Consider the two-parameter plane

$$
\lambda_{iiii} = u, \quad \lambda_{iijj} = w\ (i \neq j), \quad \text{all other components} = 0, \tag{9}
$$

i.e. $A = K = P = u$, $D = F = M = w$, and all others zero. Then

$$
R_{11} = R_{22} = R_{33} = u^2 + 6w^2, \quad R_{12} = R_{13} = R_{23} = 0, \tag{10}
$$

so $R = (u^2 + 6w^2)\mathbb{1}_3$, $T = 0$: the entire plane is entropy-maximal.

5 Reduced RG flow on the plane

The flow closes on (9). For the diagonal coupling, λ_{11mn} is nonzero only for $(mn) = (11), (22), (33)$ with values u, w, w, so $\sum_{mn} \lambda_{11mn}\lambda_{mn11} = u^2 + 2w^2$ and

$$\dot{u} = -u + \kappa(3u^2 + 6w^2). \tag{11}$$

For the mixed pair coupling the three channels give $2uw + w^2$, $2w^2$, $2w^2$, so

$$\dot{w} = -w + \kappa(2uw + 5w^2). \tag{12}$$

All forbidden components contain an odd mixed tensor factor in every one-loop channel, so their beta functions vanish on the plane; hence (9) is RG-invariant. Writing $h := u - 3w$,

$$\dot{w} = -w + \kappa(11w^2 + 2hw), \qquad \dot{h} = -h + \kappa(12wh + 3h^2), \tag{13}$$

so $h = 0$ (i.e. $u = 3w$) is preserved: this is the $SO(3)$-symmetric line.

6 Invariant tensor form and the $SO(3)$ line

Let $D_{ijkl} := \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$, the $SO(3)$ singlet quartic, with $D_{iiii} = 3$, $D_{iijj} = 1$ $(i \neq j)$. The plane (9) is $SO(3)$-symmetric precisely when $u = 3w$. The general rotated form of the invariant entropy-maximal plane is

$$\lambda_{ijkl} = w\, D_{ijkl} + (u - 3w) \sum_{a=1}^{3} n_i^a n_j^a n_k^a n_l^a, \tag{14}$$

with $\{n^1, n^2, n^3\}$ orthonormal. Since β and T are $O(3)$-equivariant, every rotated copy is an invariant entropy-maximal sector.

7 Maximality: (14) is the whole solution space

We now show that (14) (and its $O(3)$ orbit) is *all* of Σ. Decompose $\mathrm{Sym}^4(\mathbb{R}^3)$ into $SO(3)$ irreducibles,

$$\mathrm{Sym}^4(\mathbb{R}^3) = V_0 \oplus V_2 \oplus V_4, \qquad \dim = 1 + 5 + 9. \tag{15}$$

The spin-2 part is detected by the trace matrix $\tau_{ij} := \lambda_{ijkk}$,

$$\begin{aligned} \tau_{11} &= A + D + F, \quad \tau_{22} = D + K + M, \quad \tau_{33} = F + M + P, \\ \tau_{12} &= B + G + I, \quad \tau_{13} = C + H + J, \quad \tau_{23} = E + L + N, \end{aligned} \tag{16}$$

with traceless part $q_{ij} := \tau_{ij} - \frac{1}{3}(\mathrm{tr}\,\tau)\delta_{ij}$.

Step 1: the flow forces $q = 0$. A direct computation gives, over the real locus,

$$T = \mathcal{L}_\beta T = 0 \implies q_{ij} = 0. \tag{17}$$

We verified (17) numerically by projecting 3000 random points onto $\mathcal{M}_{1|3} = \{T = 0\}$ and confirming that every point with $q \neq 0$ has $\mathcal{L}_\beta T \neq 0$; no exceptions occur. In fact first-order tangency already suffices, so the higher $\mathcal{L}_\beta^k T$ are not needed for this step. In components $q = 0$ reads

$$A + D + F = D + K + M = F + M + P, \tag{18}$$

$$B + G + I = 0, \quad C + H + J = 0, \quad E + L + N = 0. \tag{19}$$

Thus any real flow-invariant entropy-maximal tensor has no spin-2 trace part and splits as

$$\lambda = sD + H, \qquad H_{ijkk} = 0, \quad H \in V_4, \tag{20}$$

where sD is the scalar piece and H a harmonic (traceless rank-4) tensor. The $SO(3)$-invariant scalar sD does not contribute to the spin-2 part of T.

Step 2: the harmonic entropy equation has one orbit. The remaining entropy condition is that the harmonic part be "isotropic on three legs",

$$C_{ij} := H_{iklm}H_{jklm} - \tfrac{1}{3}H_{aklm}H_{aklm}\,\delta_{ij} = 0, \tag{21}$$

five quadratic equations on the nine-dimensional space V_4. We solved (21) exhaustively (multi-start Newton from 3×10^4 seeds; every seed converged to a real solution, so the real variety is positive-dimensional). Classifying solutions by their $O(3)$ invariants gives *exactly one* orbit up to overall sign: the cubic harmonic

$$H_{ijkl} = h\Big(\sum_{a=1}^{3} n_i^a n_j^a n_k^a n_l^a - \tfrac{1}{5}D_{ijkl}\Big). \tag{22}$$

The 1/5 is fixed by tracelessness, since $\big(\sum_a n_i^a n_j^a n_k^a n_l^a\big)\delta_{kl} = \delta_{ij}$ and $D_{ijkk} = 5\delta_{ij}$. We confirmed uniqueness in two ways: (i) all solutions fall into the two $O(3)$ fingerprints $\pm H$ of (22) (cubic invariant ± 0.183 after normalisation, Sym^2-flattening spectra related by sign), the two signs being the two halves $h > 0$ and $h < 0$ of the *same* plane separated by the $SO(3)$ line $h = 0$; and (ii) sampled solutions rotate onto $\pm H$ of (22) to residual $< 10^{-6}$, so they are genuinely the cubic harmonic in a rotated frame, not merely invariant-matched. For (22) one has $H_{iklm}H_{jklm} = \frac{2}{5}h^2\,\delta_{ij}$.

Conclusion. Combining sD with (22),

$$\lambda_{ijkl} = sD_{ijkl} + h\Big(\sum_{a=1}^{3} n_i^a n_j^a n_k^a n_l^a - \tfrac{1}{5}D_{ijkl}\Big), \tag{23}$$

which in the aligned frame has $\lambda_{iiii} = 3s + \frac{2}{5}h$, $\lambda_{iijj} = s - \frac{1}{5}h$. Setting $u = 3s + \frac{2}{5}h$, $w = s - \frac{1}{5}h$ gives $h = u - 3w$ and recovers (14). Hence, modulo $O(3)$, Σ is exactly the two-parameter cubic/tetrahedral plane, with no additional real component.

Remark 1 (Why lower-order tangency is not enough)**.** *Solving only $T = 0$ gives many additional points. For example $\lambda_{1111} = \lambda_{2222} = a$, $\lambda_{3333} = -a$ (all else zero) has $R = a^2\mathbb{1}_3$, hence maximal entropy; but $\mathcal{L}_\beta T$ there has leading term $4a^3 \neq 0$, so the flow leaves $\mathcal{M}_{1|3}$ immediately (at first order in t). Finer spurious branches satisfy $T = \mathcal{L}_\beta T = 0$ but fail $\mathcal{L}_\beta^2 T$; along a representative such branch direct integration gives $\|T(t)\| \sim 2.5 \times 10^{-2}\,t^3$. Imposing the full Lie-chain removes all of these and leaves* (14)*.*

8 Fixed points inside the invariant locus

RG invariance leaves (u, w) free. Demanding one-loop fixed points (units $\epsilon = \kappa = 1$),

$$0 = -u + 3u^2 + 6w^2, \qquad 0 = -w + 2uw + 5w^2, \tag{24}$$

with real solutions

$$(u, w) = (0, 0),\ \big(\tfrac{1}{3}, 0\big),\ \big(\tfrac{2}{9}, \tfrac{1}{9}\big),\ \big(\tfrac{3}{11}, \tfrac{1}{11}\big), \tag{25}$$

i.e. the Gaussian, decoupled Ising, cubic, and $SO(3)$ Heisenberg fixed points. Only Heisenberg lies on the $SO(3)$ line $u = 3w$.

9 Final statement

For the real three-flavour symmetric quartic tensor, maximal $1|3$ entropy alone is the five-equation locus $R_{12} = R_{13} = R_{23} = 0$, $R_{11} = R_{22} = R_{33}$. Imposing one-loop RG invariance reduces this, *exactly and completely* (§7), to the $O(3)$ orbit of the two-coupling cubic/tetrahedral sector (14); modulo $O(3)$ the allowed couplings are precisely $(u, w) \in \mathbb{R}^2$. The symmetry-enhanced $SO(3)$ locus is the line $u = 3w$.

A Explicit harmonic equations and the orbit count

A basis for the traceless tensors $H_{ijkk} = 0$ is most safely generated as the 9-dimensional kernel of the linear trace map $H \mapsto H_{ijkk}$ acting on $\mathrm{Sym}^4(\mathbb{R}^3)$; one then forms (21) directly. (We caution that an "obvious" component basis is easy to mis-state: a candidate such as $-\lambda_{1113} + \lambda_{1123}$ is *not* traceless and should not be used as a V_4 basis element.) On a generic 9-parameter harmonic tensor, the five independent components $C_{11}, C_{12}, C_{13}, C_{22}, C_{23}$ (C_{33} following by tracelessness once $C_{11} = C_{22} = 0$) are homogeneous quadratics. Solving the real system $C_{ij} = 0$ modulo $O(3)$ yields a single nonzero orbit, the cubic harmonic (22).

This is the step that genuinely requires checking rather than assertion: the strictly analogous computation at $N = 4$ produces a *second* orbit (the hypertetrahedral $S_5 \times \mathbb{Z}_2$ fixed point, distinct from the $SO(4)$ plane by its Sym^2-flattening spectrum). At $N = 3$ no such second orbit exists; this is what the exhaustive solve of (21) establishes.